Consider a uniform solid sphere of mass M and radius a moving on a rough horizontal surface with coefficient of kinetic friction μ. We take the forward direction of the centre of mass as positive, and choose the positive angular velocity so that rolling to the right corresponds to $v = a\omega$.

(1)

The moment of inertia about an axis passing through the centre is

$$I = \int r_\perp^2 \, dm.$$

Using spherical coordinates, where $r_\perp = r\sin\theta$ and $dm = \rho \, dV$, we obtain

$$\begin{aligned} I &= \rho \int_0^a r^4 \, dr \int_0^{2\pi} d\phi \int_0^{\pi} \sin^3\theta \, d\theta \\ &= \rho \cdot \frac{a^5}{5} \cdot 2\pi \cdot \frac{4}{3}. \end{aligned}$$

Since the mass density is

$$\rho = \frac{M}{\frac{4}{3}\pi a^3} = \frac{3M}{4\pi a^3},$$

it follows that

$$\boxed{I = \frac{2}{5}Ma^2}.$$

(2)

The sphere is struck through its centre and initially moves with $v(0) = v_0$ and $\omega(0) = 0$.

(a)

While the sphere is slipping forward, kinetic friction acts backwards with magnitude μMg. This force slows the centre of mass, but its torque increases the angular velocity in our positive convention. Thus,

$$\boxed{M\frac{dv}{dt} = -\mu Mg}, \qquad \boxed{I\frac{d\omega}{dt} = \mu Mga}.$$

(b)

The translational equation gives

$$\frac{dv}{dt} = -\mu g \quad \Longrightarrow \quad \boxed{v(t) = v_0 - \mu g t}.$$

Using $I = \frac{2}{5}Ma^2$ in the rotational equation,

$$\frac{d\omega}{dt} = \frac{\mu Mga}{I} = \frac{5\mu g}{2a}.$$

With $\omega(0) = 0$,

$$\boxed{\omega(t) = \frac{5\mu g}{2a}\, t}.$$

These expressions hold during the slipping stage, $0 \le t \le T$.

(c)

The slip velocity at the contact point is

$$\begin{aligned} u(t) &= v(t) - a\omega(t) \\ &= v_0 - \mu g t - \frac{5}{2}\mu g t \\ &= v_0 - \frac{7}{2}\mu g t. \end{aligned}$$

Pure rolling begins at $t = T$, when $u(T) = 0$. Hence

$$\boxed{T = \frac{2v_0}{7\mu g}}.$$

The centre-of-mass velocity at that instant is

$$v(T) = v_0 - \mu g T = v_0 - \frac{2}{7}v_0,$$

so that

$$\boxed{v(T) = \frac{5}{7}v_0}.$$

The corresponding angular velocity is

$$\boxed{\omega(T) = \frac{v(T)}{a} = \frac{5v_0}{7a}}.$$

(d)

Once pure rolling begins, no frictional force is needed to maintain the motion on the horizontal surface. Therefore both velocities remain constant for $t > T$:

$$\boxed{v(t) = \frac{5}{7}v_0}, \qquad \boxed{\omega(t) = \frac{5v_0}{7a}}.$$

(e)

The initial kinetic energy is purely translational:

$$K_0 = \frac{1}{2}Mv_0^2.$$

At $t = T$, the sphere possesses both translational and rotational kinetic energy:

$$\begin{aligned} K_T &= \frac{1}{2}M\left(\frac{5}{7}v_0\right)^2 + \frac{1}{2}\left(\frac{2}{5}Ma^2\right)\left(\frac{5v_0}{7a}\right)^2 \\ &= \frac{5}{14}Mv_0^2. \end{aligned}$$

The total work done by kinetic friction is the change in kinetic energy:

$$W_f = K_T - K_0 = -\frac{1}{7}Mv_0^2.$$

Therefore, the *magnitude* of the work is

$$\boxed{|W_f| = \frac{1}{7}Mv_0^2}.$$

Note. Friction does work through the displacement of the *contact point*, not simply through the displacement of the centre of mass. Equivalently,

$$W_f = -\mu Mg\int_0^T u(t)\,dt = -\mu Mg\int_0^T \left(v_0 - \frac{7}{2}\mu g t\right)dt = -\frac{1}{7}Mv_0^2.$$

(3)

Now a horizontal impulse of magnitude P is applied at a height h above the centre of the sphere.

(a)

Linear impulse gives

$$Mv_1 = P \quad\Longrightarrow\quad \boxed{v_1 = \frac{P}{M}}.$$

Taking angular impulse about the centre,

$$I\omega_1 = Ph \quad\Longrightarrow\quad \boxed{\omega_1 = \frac{Ph}{I} = \frac{5Ph}{2Ma^2}}.$$

(b)

For the sphere to roll without slipping immediately after impact, we require $v_1 = a\omega_1$. Therefore,

$$\frac{P}{M} = a\frac{Ph_1}{I} \quad\Longrightarrow\quad h_1 = \frac{I}{Ma}.$$

Using the moment of inertia found in (1),

$$\boxed{h_1 = \frac{2}{5}a}.$$

(4)

The impulse is now applied at a distance ℓ *below* the centre, where $0 < \ell < a$. It initially produces backspin.

(a)

The initial velocities are

$$v(0) = \frac{P}{M}, \qquad \omega(0) = -\frac{P\ell}{I}.$$

Since

$$u(0) = v(0) - a\omega(0) = \frac{P}{M} + \frac{aP\ell}{I} > 0,$$

friction initially acts backwards, just as in part (2). Therefore, during slipping,

$$v(t) = \frac{P}{M} - \mu g t, \qquad \omega(t) = -\frac{P\ell}{I} + \frac{5\mu g}{2a} t.$$

The slip velocity is

$$\begin{aligned} u(t) &= v(t) - a\omega(t) \\ &= \frac{P}{M} + \frac{aP\ell}{I} - \frac{7}{2}\mu g t \\ &= \frac{P}{M}\left(1 + \frac{5\ell}{2a}\right) - \frac{7}{2}\mu g t. \end{aligned}$$

Setting $u(T) = 0$ gives

$$\boxed{T = \frac{2P}{7\mu M g}\left(1 + \frac{5\ell}{2a}\right)}.$$

Hence the centre-of-mass velocity when pure rolling begins is

$$\begin{aligned} v(T) &= \frac{P}{M} - \mu g T \\ &= \frac{P}{M}\left[1 - \frac{2}{7}\left(1 + \frac{5\ell}{2a}\right)\right] \\ &= \boxed{\frac{5P}{7M}\left(1 - \frac{\ell}{a}\right)}. \end{aligned}$$

(b)

Because $0 < \ell < a$,

$$1 - \frac{\ell}{a} > 0 \quad \Longrightarrow \quad v(T) = \frac{5P}{7M}\left(1 - \frac{\ell}{a}\right) > 0.$$

The translational velocity decreases monotonically during slipping, but it remains positive until pure rolling begins. After that, it stays constant and positive. Consequently, the sphere cannot reverse its direction and return to its starting point:

$$\boxed{\text{Returning to the initial position is impossible for } 0 < \ell < a.}$$